Code No: 155CX **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
PRINCIPLES OF PROGRAMMING LANGUAGES
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Describe the criteria of success for a good programming language.
b) What is the difference between a phase and a pass of a compiler? Under what circumstances do a compiler have multiple phases? [7+8]

2.a) Exception handling is very important, but often neglected by programming languages. Comment on it.
b) Give the attribute grammar for a simple assignment statement.
c) Write a denotation semantics mapping function for "for" statement in Ada. [5+5+5]

3.a) Explain about static binding and dynamic binding with its relative merits and demerits.
b) What are the design issues of logically controlled loop statements? Explain briefly. [8+7]

4.a) What is meant by type compatibility? Explain with an example.
b) Discuss the design and implementation issues of pointer data types. [7+8]

5.a) Describe pass-by-value and pass-by-result parameter passing methods with examples.
b) What are the distinct semantic models for formal parameters? Explain. [8+7]

6. List and explain various functional characteristics of OOP in Small Talk and C++. [15]

7. Explain exception propagation and exception handling in C++ with illustrative examples. [15]

8.a) Write down the applications of logic programming.
b) Explain the pragmatics of scripting languages. [7+8]

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